Q1 - 24 January - Shift 1

The value of $12\int_0^3 \left|x^2-3x+2\right| dx$ is ________

Space for your notes:

Q2 - 24 January - Shift 1

The value of $\frac{8}{\pi}\int_0^{\frac{\pi}{2}} \frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}} dx$ is ________.

Space for your notes:

Q3 - 24 January - Shift 2

$$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}} dx \text{ is equal to}$$

Space for your notes:

(1) $\frac{\pi}{3}$

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{6}$

(4) 2π

Q4 - 24 January - Shift 2

Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$ such that $f(x) > 0$ and

$$f(x) + \int_0^x f(t)\sqrt{1-(\log_e f(t))^2}\, dt = e, \ \forall x \in \left[0, \frac{\pi}{2}\right].$$

Then $\left(6 \log_e f\left(\frac{\pi}{6}\right)\right)^2$ is equal to ________.

Space for your notes:

Q5 - 25 January - Shift 1

The minimum value of the function $f(x) = \int_0^2 e^{|x-t|}\, dt$ is

(1) $2(e-1)$ (2) $2e-1$

(3) 2 (4) $e(e-1)$

Space for your notes:

Q6 - 25 January - Shift 1

The value of

$$\lim_{n\to\infty}\frac{1+2-3+4+5-6+\ldots+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$$

is :

(1) $\frac{\sqrt{2}+1}{2}$

(2) $3(\sqrt{2}+1)$

(3) $\frac{3}{2}(\sqrt{2}+1)$

(4) $\frac{3}{2\sqrt{2}}$

Space for your notes:

Q7 - 25 January - Shift 2

The integral $16\int_1^2 \frac{dx}{x^3\left(x^2+2\right)^2}$ is equal to

(1) $\frac{11}{6}+\log_e 4$ (2) $\frac{11}{12}+\log_e 4$

(3) $\frac{11}{12}-\log_e 4$ (4) $\frac{11}{6}-\log_e 4$

Space for your notes:

Q8 - 25 January - Shift 2

If $\int_{\frac{1}{3}}^{3}|\log_e x|dx = \frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$, where m and n are coprime natural numbers, then m^2+n^2-5 is equal to ______.

Space for your notes:

Q9 - 29 January - Shift 1

Let [x] denote the greatest integer $\leq x$. Consider the function $f(x) = \max\{x^2, 1+[x]\}$. Then the value of the integral $\int_0^2 f(x)dx$ is :

(1) $\frac{5+4\sqrt{2}}{3}$ (2) $\frac{8+4\sqrt{2}}{3}$

(3) $\frac{1+5\sqrt{2}}{3}$ (4) $\frac{4+5\sqrt{2}}{3}$

Space for your notes:

Q10 - 29 January - Shift 1

Let $f(x) = x + \frac{a}{\pi^2 - 4}\sin x + \frac{b}{\pi^2 - 4}\cos x$, $x \in \mathbb{R}$ be a function which satisfies $f(x) = x + \int_0^{\pi/2} \sin(x+y) f(y)dy$. Then (a + b) is equal to

(1) $-\pi(\pi+2)$ (2) $-2\pi(\pi+2)$

(3) $-2\pi(\pi-2)$ (4) $-\pi(\pi-2)$

Space for your notes:

Q11 - 29 January - Shift 2

The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ is equal to

(1) $\pi \log_e 2$ (2) $\frac{1}{2}\log_e 2$

(3) $\frac{\pi}{4}\log_e 2$ (4) $\frac{\pi}{2}\log_e 2$

Space for your notes:

Q12 - 29 January - Shift 2

The value of the integral $\int_{1}^{2}\left(\frac{t^4+1}{t^6+1}\right)dt$ is

(1) $\tan^{-1}\frac{1}{2}+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

(2) $\tan^{-1}2-\frac{1}{3}\tan^{-1}8+\frac{\pi}{3}$

(3) $\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

(4) $\tan^{-1}\frac{1}{2}-\frac{1}{3}\tan^{-1}8+\frac{\pi}{3}$

Space for your notes:

Q13 - 30 January - Shift 1

If [t denotes the greatest integer ≤ 1, then the value of $\frac{3(e-1)^2}{e}\int_{1}^{2} x^2 e^{[x]+[x^3]}dx$ is :

(1) $e^9 - e$ (2) $e^8 - e$

(3) $e^7 - 1$ (4) $e^8 - 1$

Space for your notes:

Q14 - 30 January - Shift 1

$\lim_{x\to 0}\frac{48}{x^4}\int_{0}^{x}\frac{t^3}{t^6+1}dt$ is equal to ______

Space for your notes:

Q15 - 30 January - Shift 2

So, it is true for every natural no. 'a_1'

$$\lim_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+\ldots+\left(3-\frac{1}{n}\right)^2\right\}$$

is equal to

(1) 12 (2) $\frac{19}{3}$

(3) 0 (4) 19

Space for your notes:

Q16 - 31 January - Shift 1

The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\frac{(2+3\sin x)}{\sin x(1+\cos x)}dx$ is equal to

(1) $\frac{7}{2}-\sqrt{3}-\log_e\sqrt{3}$

(2) $-2+3\sqrt{3}+\log_e\sqrt{3}$

(3) $\frac{10}{3}-\sqrt{3}+\log_e\sqrt{3}$

(4) $\frac{10}{3}-\sqrt{3}-\log_e\sqrt{3}$

Space for your notes:

Q17 - 31 January - Shift 1

Let a differentiable function f satisfy $f(x)+\int_3^x\frac{f(t)}{t}dt=\sqrt{x+1}, x\geq 3$. Then 12f(8) is equal to:

(1) 34

(2) 19

(3) 17

(4) 1

Space for your notes:

Q18 - 31 January - Shift 1

Let $\alpha \in (0, 1)$ and $\beta = \log_e(1 - \alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + ... + \frac{x^n}{n}$, $x \in (0, 1)$.

Then the integral $\int_0^\alpha \frac{t^{50}}{1-t} dt$ is equal to

(1) $\beta - P_{50}(\alpha)$

(2) $-(\beta + P_{50}(\alpha))$

(3) $P_{50}(\alpha) - \beta$

(4) $\beta + P_{50}(\alpha)$

Space for your notes:

Q19 - 31 January - Shift 2

Let $\alpha > 0$. If $\int_0^\alpha \frac{x}{\sqrt{x+\alpha} - \sqrt{x}} dx = \frac{16 + 20\sqrt{2}}{15}$, then α is equal to :

(1) 2

(2) 4

(3) $\sqrt{2}$

(4) $2\sqrt{2}$

Space for your notes:

Q20 - 31 January - Shift 2

If $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt,\ \ x>0,$

then $\phi'\left(\frac{\pi}{4}\right)$ is equal to :

(1) $\frac{8}{\sqrt{\pi}}$

(2) $\frac{4}{6+\sqrt{\pi}}$

(3) $\frac{8}{6+\sqrt{\pi}}$

(4) $\frac{4}{6-\sqrt{\pi}}$

Space for your notes:

Q21 - 01 February - Shift 1

$\lim_{n\to\infty}\left(\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+\ldots+\frac{1}{2n}\right)$ is equal to :-

(1) 0

(2) $\log_e 2$

(3) $\log_e\left(\frac{3}{2}\right)$

(4) $\log_e\left(\frac{2}{3}\right)$

Space for your notes:

Q22 - 01 February - Shift 1

If

$$\int_0^1 (x^{21}+x^{14}+x^7)(2x^{14}+3x^7+6)^{1/7}\,dx=\frac{1}{l}(11)^{m/n}$$

where l, m, $n \in \mathbb{N}$, m and n are coprime then $l+m+n$ is equal to ____.

Space for your notes:

Q23 - 01 February - Shift 2

The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x} dx$ is

Space for your notes:

(1) $\frac{\pi^2}{6}$

(2) $\frac{\pi^2}{12\sqrt{3}}$

(3) $\frac{\pi^2}{3\sqrt{3}}$

(4) $\frac{\pi^2}{6\sqrt{3}}$

Q24 - 01 February - Shift 2

If $\int_0^{\pi} \frac{5^{\cos x}\left(1+\cos x \cos 3x+\cos^2 x+\cos^3 x \cos 3x\right) dx}{1+5^{\cos x}} = \frac{k\pi}{16}$,
then k is equal to ______.

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (22)	**Q2** (2)	**Q3** (4)	**Q4** (27)
Q5 (1)	**Q6** (3)	**Q7** (4)	**Q8** (20)
Q9 (1)	**Q10** (2)	**Q11** (4)	**Q12** (3)
Q13 (2)	**Q14** (12)	**Q15** (4)	**Q16** (3)
Q17 (3)	**Q18** (2)	**Q19** (1)	**Q20** (3)
Q21 (2)	**Q22** (63)	**Q23** (4)	**Q24** (26)

Q1 (22)

$$12\int_0^3 \left|x^2-3x+2\right| dx$$

$$=12\int_0^3 \left|\left(x-\frac{3}{2}\right)^2-\frac{1}{4}\right| dx$$

If $x-\dfrac{3}{2}=t$

$dx = dt$

$$=24\int_0^{3/2} \left|t^2-\frac{1}{4}\right| dt$$

$$=24\left[-\int_0^{1/2}\left(t^2-\frac{1}{4}\right)dt+\int_{1/2}^{3/2}\left(t^2-\frac{1}{4}\right)dt\right]=22$$

Q2 (2)

$$I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx \;\ldots\ldots\ldots\ldots(1)$$

Using $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

$$I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}\frac{(\sin x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx \;\ldots\ldots\ldots\ldots(2)$$

Adding (1) & (2)

$$2I=\frac{8}{\pi}\int_0^{\frac{\pi}{2}}1\,dx$$

$I=2$

Q3 (4)

$$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}\,dx$$

We have $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$

Hence $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}\,dx = \frac{48}{2} \times \left[\sin^{-1}\frac{2x}{3}\right]_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}}$

$$= 24 \times \left[\sin^{-1}\left(\frac{2}{3} \times \frac{3\sqrt{3}}{4}\right) - \sin^{-1}\left(\frac{2}{3} \times \frac{3\sqrt{2}}{4}\right)\right]$$

$$= 24 \times \left[\sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}\frac{1}{\sqrt{2}}\right]$$

$$= 24 \times \left(\frac{\pi}{3} - \frac{\pi}{4}\right)$$

$$= 24 \times \frac{\pi}{12} = 2\pi$$

Q4 (27)

$$f(x) + \int_0^x f(t)\sqrt{1-\left(\log_e f(t)\right)^2}\,dt = e$$

$$\Rightarrow f(0) = e$$

$$f'(x) + f(x)\sqrt{1-\left(\ln f(x)\right)^2} = 0$$

$$f(x) = y$$

$$\frac{dy}{dx} = -y\sqrt{1-(\ln y)^2}$$

$$\int \frac{dy}{y\sqrt{1-(\ln y)^2}} = -\int dx$$

Put $\ln y = t$

$$\int \frac{dt}{\sqrt{1-t^2}} = -x + C$$

$$\sin^{-1} t = -x + C \Rightarrow \sin^{-1}(\ln y) = -x + C$$

$$\sin^{-1}\left(\ln f(x)\right) = -x + C$$

$$f(0) = e$$

$$\Rightarrow \frac{\pi}{2} = C$$

$$\Rightarrow \sin^{-1}\left(\ln f(x)\right) = -x + \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}\left(\ln f\left(\frac{\pi}{6}\right)\right) = \frac{-\pi}{6} + \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}\left(\ln f\left(\frac{\pi}{6}\right)\right) = \frac{\pi}{3}$$

$$\Rightarrow \ln f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \text{ we need } \left(6 \times \frac{\sqrt{3}}{2}\right)^2 = 27$$

Q5 (1)

For $x \le 0$

$$f(x) = \int_0^2 e^{t-x}dt = e^{-x}\left(e^2 - 1\right)$$

For $0 < x < 2$

$$f(x) = \int_0^x e^{x-t}dt + \int_x^2 e^{t-x}dt = e^x + e^{2-x} - 2$$

For $x \ge 2$

$$f(x) = \int_0^2 e^{x-t}dt = e^{x-2}(e^2 - 1)$$

For $x \le 0$, f(x) is $\downarrow$ and $x \ge 2$, f(x) is $\uparrow$

$\therefore$ Minimum value of f(x) lies in $x \in (0, 2)$

Applying $A.M \ge G.M$,
minimum value of f(x) is $2(e - 1)$

Q6 (3)

$$\lim_{n\to\infty} \frac{0+3+6+9+\ldots..n \text{ terms}}{\sqrt{2n^4+4n+3} - \sqrt{n^4+5n+4}}$$

$$\lim_{n\to\infty} \frac{3n(n-1)}{2\left(\sqrt{2n^4+4n+3} - \sqrt{n^4+5n+4}\right)}$$

$$= \frac{3}{2\left(\sqrt{2}-1\right)} = \frac{3}{2}\left(\sqrt{2}+1\right)$$

Q7 (4)

$$I = 16\int_1^2 \frac{dx}{x^3(x^2+2)^2}$$

$$= 16\int_1^2 \frac{dx}{x^3x^4\left(1+\frac{2}{x^2}\right)^2}$$

Let, $1+\frac{2}{x^2} = t \Rightarrow \frac{-4}{x^3}dx = dt$

$$I = -4\int_3^{\frac{3}{2}} \frac{dt}{\left(\frac{2}{t-1}\right)^2 t^2}$$

$$I = -4\int_3^{\frac{3}{2}} \left(\frac{t-1}{2}\right)^2 \frac{dt}{t^2}$$

$$I = -\frac{4}{4}\int_3^{\frac{3}{2}} \left(1-\frac{2}{t}+\frac{1}{t^2}\right)dt$$

$$I = -1\left[t-2\ell n|t|-\frac{1}{t}\right]_3^{\frac{3}{2}}$$

$$I = -1\left[\left(\frac{3}{2}-2\ell n\frac{3}{2}-\frac{2}{3}\right)-\left(3-2\ell n3-\frac{1}{3}\right)\right]$$

$$I = -1\left[2\ell n2-\frac{11}{6}\right]$$

$$I = \frac{11}{6}-\ell n4$$

Q8 (20)

$$\int_{\frac{1}{3}}^{3} |\ell nx|\,dx = \int_{\frac{1}{3}}^{1} (-\ell nx)dx + \int_{1}^{3} (\ell nx)\,dx$$

$$= -[x\,\ell nx - x]_{1/3}^{1} + [x\,\ell nx - x]_{1}^{3}$$

$$= -\left[-1-\left(\frac{1}{3}\ell n\frac{1}{3}-\frac{1}{3}\right)\right]+[3\,\ell n3-3-(-1)]$$

$$= \left[-\frac{2}{3}-\frac{1}{3}\ell n\frac{1}{3}\right]+[3\,\ell n3-2]$$

$$= -\frac{4}{3}+\frac{8}{3}\ell n3$$

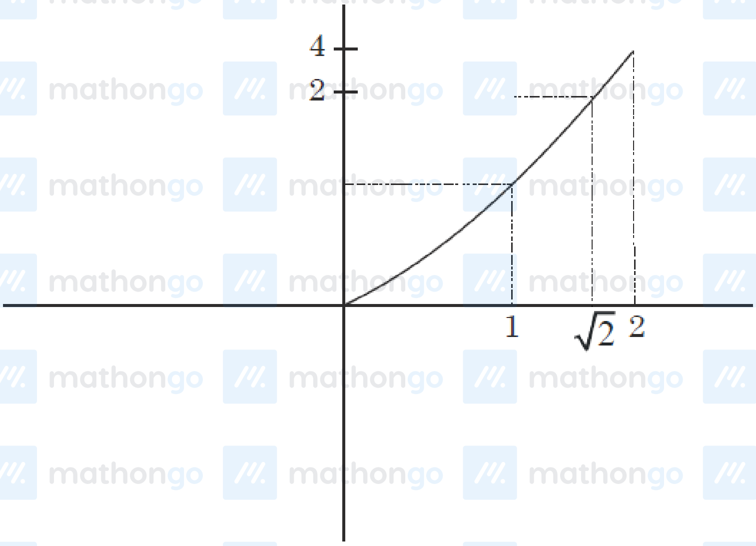
$$= \frac{4}{3}(2\ell n3-1)$$

$$= \frac{4}{3}\left(\ell n\frac{9}{e}\right)$$

$\therefore \quad m = 4,\ n = 3$

Now, $m^2 + n^2 - 5 = 16 + 9 - 5 = 20$

Q9 (1)



$$A=\int_0^1 1.dx+\int_1^{\sqrt{2}} 2dx+\int_{\sqrt{2}}^2 x^2dx$$

$$=1+2\sqrt{2}-2+\frac{8}{3}-\frac{2\sqrt{2}}{3}$$

$$=\frac{5}{3}+\frac{4\sqrt{2}}{3}$$

Q10 (2)

$$f(x)=x+\int_0^{\pi/2}(\sin x\cos y+\cos x\sin y)f(y)\,dy$$

$$f(x)=x+\int_0^{\pi/2}\left((\cos y\, f(y)dy)\sin x+(\sin y\, f(y)dy)\cos x\right)$$

...(1)

On comparing with

$$f(x)=x+\frac{a}{\pi^2-4}\sin x+\frac{b}{\pi^2-4}\cos x,\ x\in\mathbb{R} \text{ then}$$

$$\Rightarrow \frac{a}{\pi^2-4}=\int_0^{\pi/2}\cos y\, f(y)\,dy \qquad \textbf{...(2)}$$

$$\Rightarrow \frac{b}{\pi^2-4}=\int_0^{\pi/2}\sin y\, f(y)\,dy \qquad \textbf{...(3)}$$

Add (2) and (3)

$$\frac{a+b}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)f(y)\,dy \ \ldots\ldots(4)$$

$$\frac{a+b}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)f\left(\frac{\pi}{2}-y\right)dy \ \ldots(5)$$

Add (4) and (5)

$$\frac{2(a+b)}{\pi^2-4}=\int_0^{\pi/2}(\sin y+\cos y)\left(\frac{\pi}{2}+\frac{(a+b)}{\pi^2-4}(\sin y+\cos y)\right)dy$$

$$=\pi+\frac{a+b}{\pi^2-4}\left(\frac{\pi}{2}+1\right)$$

$$(a+b)=-2\pi(\pi+2)$$

Q11 (4)

$$I=\int_{1/2}^{2}\frac{\tan^{-1}x}{x}dx \quad \ldots\ldots \text{(i)}$$

Put $x=\frac{1}{t} \quad dx=-\frac{1}{t^2}dt$

$$I=-\int_{2}^{1/2}\frac{\tan^{-1}\frac{1}{t}}{\frac{1}{t}}\cdot\frac{1}{t^2}dt=-\int_{2}^{1/2}\frac{\tan^{-1}\frac{1}{t}}{t}dt$$

$$I=\int_{1/2}^{2}\frac{\cot^{-1}t}{t}dt=\int_{1/2}^{2}\frac{\cot^{-1}x}{x}dx \quad \ldots\ldots \text{(ii)}$$

Add both equation

$$2I=\int_{1/2}^{2}\frac{\tan^{-1}x+\cot^{-1}x}{x}dx=\frac{\pi}{2}\int_{1/2}^{2}\frac{dx}{x}=\frac{\pi}{2}(\ell n2)_{1/2}^{2}$$

$$=\frac{\pi}{2}\left(\ell n2-\ell n\frac{1}{2}\right)=\pi\ell n2$$

$$I=\frac{\pi}{2}\ell n2$$

Q12 (3)

$$I=\int_1^2\left(\frac{t^4+1}{t^6+1}\right)dt$$

$$=\int_1^2\frac{\left(t^4+1-t^2\right)+t^2}{\left(t^2+1\right)\left(t^4-t^2+1\right)}dt$$

$$=\int_1^2\left(\frac{1}{t^2+1}+\frac{t^2}{t^6+1}\right)dt$$

$$=\int_1^2\left(\frac{1}{t^2+1}+\frac{1}{3}\frac{3t^2}{(t^3)^2+1}\right)dt$$

$$=\tan^{-1}(t)+\frac{1}{3}\tan^{-1}(t^3)\Big|_1^2$$

$$=(\tan^{-1}(2)-\tan^{-1}(1))+\frac{1}{3}(\tan^{-1}(2^3)-\tan^{-1}(1^3))$$

$$=\tan^{-1}(2)+\frac{1}{3}\tan^{-1}(8)-\frac{\pi}{3}$$

Q13 (2)

$$\int_1^2 x^2 e^{[x^3]+1} dx$$

$x^3 = t$

$3x^2\, dx = dt$

$$= \frac{e}{3}\int_1^8 e^{[t]} dt$$

$$= \frac{e}{3}\left\{\int_1^2 e\, dt + \int_2^3 e^2\, dt + \ldots\ldots + \int_7^8 e^7\, dt\right\}$$

$$= \frac{e}{3}(e + e^2 + \ldots\ldots + e^7)$$

$$= \frac{e^2}{3}(1 + e + \ldots\ldots + e^6) = \frac{e^2}{3}\frac{(e^7 - 1)}{(e-1)}$$

$$\frac{3(e-1)}{e}\int_1^2 x^2 \times e^{[x]+[x^3]} dx = \frac{3}{e}(e-1) \times \frac{e^2}{3}\frac{(e^7-1)}{(e-1)}$$

$$= e(e^7 - 1)$$

$$= e^8 - e$$

Q14 (12)

$$48 \lim_{x\to 0} \frac{\int_0^x \frac{t^3}{t^6+1} dt}{x^4} \quad \left(\frac{0}{0}\right)$$

Applying L' Hospitals Rule

$$48 \lim_{x\to 0} \frac{x^3}{x^6+1} \times \frac{1}{4x^3}$$

$= 12$

Q15 (4)

$$\lim_{n\to\infty}\frac{3}{n}\sum_{r=0}^{n-1}\left(2+\frac{r}{n}\right)^2$$

$$=3\int_0^1(2+x)^2\,dx=27-8=19$$

Q16 (3)

$$\int_{\pi/3}^{\pi/2}\left(\frac{2+3\sin x}{\sin x(1+\cos x)}\right)dx=2\int_{\pi/3}^{\pi/2}\frac{dx}{\sin x+\sin x\cos x}+3$$

$$3\int_{\pi/3}^{\pi/2}\frac{dx}{1+\cos x}$$

$$\int_{\pi/3}^{\pi/2}\frac{dx}{1+\cos x}=\int_{\pi/3}^{\pi/2}\frac{1-\cos x}{\sin^2 x}dx$$

$$=\int_{\pi/3}^{\pi/2}\left(\operatorname{cosec}^2 x-\cot x\operatorname{cosec} x\right)dx$$

$$=\left(\operatorname{cosec} x-\cot x\right)\Big|_{\pi/3}^{\pi/2}=(1)-\left(\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}\right)=1-\frac{1}{\sqrt{3}}$$

$$\int_{\pi/3}^{\pi/2}\frac{dx}{\sin x(1+\cos x)}=$$

$$\int\frac{dx}{(2\tan x/2)\left(1+1-\tan^2 x/2\right)}$$

$$=\int\frac{\left(1+\tan^2 x/2\right)\sec^2 x/2\,dx}{2\tan x/2\;2}$$

$$\tan x/2=t \qquad \sec x/2\,\frac{1}{2}dx=dt$$

$$\frac{1}{2}\int\left(\frac{1+t^2}{t}\right)dt=\frac{1}{2}\left[\ell n t+\frac{t^2}{2}\right]_{\frac{1}{\sqrt{3}}}^{1}$$

$$=\frac{1}{2}\left[\left(0+\frac{1}{2}\right)-\left(\ell n\frac{1}{\sqrt{3}}+\frac{1}{6}\right)\right]=\left(\frac{1}{3}+\ell n\sqrt{3}\right)\frac{1}{2}$$

$$=\left(\frac{1}{6}+\frac{1}{2}\ell n\sqrt{3}\right)$$

$$2\left(\frac{1}{6}+\frac{1}{2}\ell n\sqrt{3}\right)+3\left(1-\frac{1}{\sqrt{3}}\right)$$

$$=\frac{1}{3}+\ell n\sqrt{3}+3-\sqrt{3}=\frac{10}{3}+\ell n\sqrt{3}-\sqrt{3}$$

Q17 (3)

Differentiate w.r.t. x

$$f'(x)+\frac{f(x)}{x}=\frac{1}{2\sqrt{x+1}}$$

$$\text{I.F.} = e^{\int \frac{1}{x}dx}=e^{\ln x}=x$$

$$xf(x)=\int \frac{x}{2\sqrt{x+1}}dx$$

$x + 1 = t^2$

$$=\int \frac{t^2-1}{2t}2tdt$$

$$xf(x)=\frac{t^3}{3}-t+c$$

$$xf(x)=\frac{(x+1)^{3/2}}{3}-\sqrt{x+1}+c$$

Also putting x = 3 in given equation $f(3)+0=\sqrt{4}$

f(3) = 2

$$\Rightarrow C=8-\frac{8}{3}=\frac{16}{3}$$

$$f(x)=\frac{\frac{(x+1)^{3/2}}{3}-\sqrt{x+1}+\frac{16}{3}}{x}$$

$$f(8)=\frac{9-3+\frac{16}{3}}{8}=\frac{34}{24}$$

$\Rightarrow$ 12 f(8) = 17

Q18 (2)

$$\int_0^{\alpha}\frac{t^{50}-1+1}{1-t}=-\int_0^{\alpha}\left(1+t+\ldots..+t^{49}\right)+\int_0^{\alpha}\frac{1}{1-t}dt$$

$$=-\left(\frac{\alpha^{50}}{50}+\frac{\alpha^{49}}{49}+\ldots..+\frac{\alpha^{1}}{1}\right)+\left(\frac{\ln(1-f)}{-1}\right)_0^{\alpha}$$

$$=-P_{50}(\alpha)-\ln(1-\alpha)`$$

$$=-P_{50}(\alpha)-\beta$$

Q19 (1)

After rationalising

$$\int_0^{\alpha} \frac{x}{\alpha}\left(\sqrt{x+\alpha}+\sqrt{x}\right)$$

$$\int_0^{\alpha} \frac{1}{\alpha}\left[(x+\alpha)^{3/2}-\alpha(x+\alpha)^{1/2}+x^{3/2}\right]$$

$$\frac{1}{\alpha}\left[\frac{2}{5}(x+\alpha)^{5/2}-\alpha\frac{2}{3}(x+\alpha)^{3/2}+\frac{2}{5}x^{5/2}\right]\Big|_0^{\alpha}$$

$$=\frac{1}{\alpha}\left(\frac{5}{2}(2\alpha)^{5/2}-\frac{2\alpha}{3}(2\alpha)^{3/2}+\frac{2}{5}\alpha^{5/2}-\frac{2}{5}\alpha^{5/2}+\frac{2}{3}\alpha^{5/2}\right)$$

$$=\frac{1}{\alpha}\left(\frac{2^{7/2}\alpha^{5/2}}{5}\frac{2^{5/2}\alpha^{5/2}}{3}+\frac{2}{3}\alpha^{5/2}\right)$$

$$=\alpha^{3/2}\left(\frac{2^{7/2}}{5}-\frac{2^{5/2}}{3}+\frac{2}{3}\right)$$

$$=\frac{\alpha^{3/2}}{15}\left(24\sqrt{2}-20\sqrt{2}+10\right)=\frac{\alpha^{3/2}}{15}\left(4\sqrt{2}+10\right)$$

Now,

$$\frac{\alpha^{3/2}}{15}\left(4\sqrt{2}+10\right)=\frac{16+20\sqrt{2}}{15}$$

$\Rightarrow \alpha = 2$

Q20 (3)

$$\phi'(x)=\frac{1}{\sqrt{x}}\left[\left(4\sqrt{2}\sin x-3\phi'(x)\right).1-0\right]-\frac{1}{2}x^{-3/2}$$

$$\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt,$$

$$\phi'\left(\frac{\pi}{4}\right)=\frac{2}{\sqrt{\pi}}\left[4-3\phi'\left(\frac{\pi}{4}\right)\right]+0$$

$$\left(1+\frac{6}{\sqrt{\pi}}\right)\phi'\left(\frac{\pi}{4}\right)=\frac{8}{\sqrt{\pi}}$$

$$\phi'\left(\frac{\pi}{4}\right)=\frac{8}{\sqrt{\pi}+6}$$

Q21 (2)

$$\lim_{n\to\infty}\left(\frac{1}{1+n}+\ldots+\frac{1}{n+n}\right)=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n+r}$$

$$=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\left(\frac{1}{1+\frac{r}{n}}\right)$$

$$=\int_0^1\frac{1}{1+x}dx=[\ell n(1+x]_0^1=\ell n2$$

Q22 (63)

$$\int\left(x^{20}+x^{13}+x^{6}\right)\left(2x^{21}+3x^{14}+6x^{7}\right)^{1/7}dx$$

$2x^{21}+3x^{14}+6x^{7}=t$

$42(x^{20}+x^{13}+x^{6})\,dx=dt$

$$\frac{1}{42}\int_{0}^{11}t^{\frac{1}{7}}dt=\left(\frac{t^{\frac{8}{7}}}{\frac{8}{7}}\times\frac{1}{42}\right)_{0}^{11}$$

$$=\frac{1}{48}\left(t^{\frac{8}{7}}\right)_{0}^{11}=\frac{1}{48}(11)^{8/7}$$

$l=48$, $m=8$, $n=7$

$l+m+n=63$

Q23 (4)

$$I=\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}}\frac{x+\frac{\pi}{4}}{2-\cos 2x}dx \qquad (1)$$

$x \to -x$

$$I=\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}}\frac{-x+\frac{\pi}{4}}{2-\cos 2x}dx \qquad (2)$$

$(1)+(2)$

$$2I=\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}}\frac{\frac{\pi}{2}}{2-\cos 2x}dx$$

$$I=\frac{\pi}{4}.2\int_{0}^{\frac{\pi}{4}}\frac{dx}{2-\cos 2x}dx$$

$$I=\frac{\pi}{4}.2\int_{0}^{\frac{\pi}{4}}\frac{\left(1+\tan^2 x\right)dx}{2\left(1+\tan^2 x\right)-\left(1-\tan^2 x\right)}$$

$$I=\frac{\pi}{4}\int_{0}^{1}\frac{dt}{3t^2+1}$$

$$\Rightarrow I=\frac{\pi}{2\sqrt{3}}\tan^{-1}\sqrt{3}$$

$$I=\frac{\pi^2}{6\sqrt{3}}$$

Q24 (26)

$$I=\int_0^{\pi}\frac{5^{\cos x}(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x)}{1+5^{\cos x}}dx$$

$$I=\int_0^{\pi}\frac{5^{-\cos x}(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x)}{1+5^{-\cos x}}dx$$

$$2I=\int_0^{\pi}(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x)dx$$

$$\cancel{2}I=\cancel{2}\int_0^{\frac{\pi}{2}}(1+\cos x\cos 3x+\cos^2 x+\cos^3 x\cos 3x)dx$$

$$I=\int_0^{\frac{\pi}{2}}(1+\sin x(-\sin 3x)+\sin^2 x-\sin^3 x\sin 3x)dx$$

$$2I=\int_0^{\frac{\pi}{2}}(3+\cos 4x+\cos^3 x\cos 3x-\sin^3 x\sin 3x)dx$$

$$2I=\int_0^{\frac{\pi}{2}}3+\cos 4x+\left(\frac{\cos 3x+3\cos x}{4}\right)\cos 3x-\sin 3x\left(\frac{3\sin x-\sin 3x}{4}\right)dx$$

$$2I=\int_0^{\frac{\pi}{2}}\left(3+\cos 4x+\frac{1}{4}+\frac{3}{4}\cos 4x\right)dx$$

$$2I=\frac{13}{4}\times\frac{\pi}{2}+\frac{7}{4}\left(\frac{\sin 4x}{4}\right)_0^{\frac{\pi}{2}}\Rightarrow I=\frac{13\pi}{16}$$